

D-Mart : Data Analysis in India

Project Scope-

Objective:

The project aims to create an SQL-based database management system for D-Mart, enabling the analysis of sales, inventory, customer data, product information, revenue, store performance, and employee salaries. This system will help track sales trends, manage stock levels, assess revenue, and streamline payroll for effective business management.

Purpose of the Project:

The D-Mart project aims to create a database management system to enhance retail operations. It will provide real-time sales insights, improve customer experiences, optimize inventory, increase revenue, track performance, manage employee records, target marketing efforts, and support strategic planning for adapting to market changes and consumer trends.

Tables

Table Name	Purpose
Customers	Stores detailed customer information.
Products	Stores product information including price and stock.
Stores	Contains D-Mart store location information.
Sales	Captures details of each sales transaction.
SalesDetails	Breaks down individual products in each sales transaction.
Inventory	Manages stock levels for products at each store.
Revenue	Tracks total revenue generated from sales.
EmployeeSalary	Stores salary details of employees.

Data Analysis Questions

Customer Analysis

1. Customer Information Retrieval: What query can be written to retrieve the customers' first names, last names, and cities of all customers? - CALL GetCustomerInformation();
2. Product Price Inquiry: How can a query be formulated to identify all products priced above 500? - CALL GetProductsAbovePrice();
3. Store Count Assessment: What query can provide the total number of stores listed in the Stores table? - CALL GetTotalStores();
4. Unique Customer Cities: How can you list the distinct cities where customers reside? - CALL GetUniqueCustomerCities();
5. Customer Sales Totals: How can a query be constructed to retrieve the total number of sales made by each customer? - CALL GetCustomerSalesTotals();

Product Analysis

1. Retrieve All Product Details: How can you retrieve all the products from the Products table using a stored procedure? - CALL GetAllProducts();
2. Count of Products in Each Category: How can you find the count of products grouped by category using a stored procedure - CALL CountProductsInEachCategory();
3. Top-N Expensive Products: How can you retrieve the top N most expensive products using a stored procedure? - CALL GetTopExpensiveProducts(5);
4. List out Products Under a Specific Price: How can you list all products that are priced under a specific amount using a stored procedure? - CALL GetProductsUnderPrice(300);
5. List out Average Price of Products: How can you find the average price of products? CALL GetAveragePrice();

Store Analysis

1. Total Stores by State: What is the total number of stores located in each state? CALL GetTotalStoresByState();
2. Cities with Highest Store Concentration: Which cities have the highest concentration of stores? - CALL GetTopCitiesByStoreCount(3);
3. Top Three Stores by Name Length: What are the top three stores based on name length or alphabetically? - CALL GetTopStoresByNameLength(3);
4. Stores Count by City: How many stores are there in each city? – CALL GetStoresByCity();

Sales Analysis

1. Total Sales Amount for Each Store: How can you list out the total sales amount for each store? - CALL GetTotalSalesByStore();
2. Customer with the Highest Total Purchases: Which customer has the highest total purchases? - CALL GetTopCustomer();
3. Average Sale Amount Across All Transactions: How can you list out the average sale amount across all transactions? - CALL GetAverageSaleAmount();

Sales Detail Analysis

1. What is the total quantity sold for each product? - CALL GetTotalQuantitySoldByProduct();
2. How can you list the total sales amount by product? - Call GetTotalSalesAmountByProduct();
3. What is the average tax amount per sale? - call GetAverageTaxAmount();
4. How can you find the total tax collected by store? - call GetTotalTaxCollectedByStore();

Inventory Analysis

1. What is the total stock quantity of all products in the inventory? - CALL GetTotalStockQuantity();
2. How many products are available in each store? - CALL GetProductCountPerStore();
3. Which product has the highest stock quantity? - CALL GetHighestStockProduct();
4. What is the product with the highest stock quantity, including its name? - CALL GetHighestStockProductwithName();

Revenue Analysis

1. How can you retrieve the total revenue generated by each store? - call GetTotalRevenueByStore();
2. What is the average revenue generated per sale? - call GetAverageRevenuePerSale();
3. How much total tax has been collected? - Call GetTotalTaxCollected();

Employee Salary

1. What is the total salary expense for the month of September 2024? - CALL GetTotalSalaryExpense();
2. Who earns the highest salary in the Sales department? - CALL GetHighestSalaryInSales();
3. Who is the highest earner in the entire department for September 2024? - CALL GetHighestEarner();
4. What is the average salary of employees in the organization? - CALL GetAverageSalary();
5. How many employees are there in each designation? - CALL CountEmployeesByDesignation();